Exercise-1

Marked Questions can be used as Revision Questions.

PART-I: SINGLE CHOICE QUESTIONS

1. If a carrier wave of 1000 kHz is used to carry the signal, the length of transmitting antenna will be equal to -
(1) 3 m (2) 30 m (3) 300 m (4) 3000 m

2. In amplitude modulation -
(1) only the amplitude is changed but frequency remains same
(2) both the amplitude and frequency change equally
(3) both the amplitude and frequency change unequally
(4) none the these

3. Modulation factor determines -
(1) only the strength of the transmitted signal (2) only the quality of the transmitted signal
(3) both the strength and quality of the signal (4) none of the above

4. Degree of modulation –
(1) can take any value (2) should be less than 100%
(3) should exceed 100% (4) none of these

5. If the maximum and minimum voltage of an AM wave are $V_{max.}$ and $V_{min.}$ respectively then modulation factor–
(1) $m = \frac{V_{max.}}{V_{max.} + V_{min.}}$ (2) $m = \frac{V_{min.}}{V_{max.} + V_{min.}}$ (3) $m = \frac{V_{max.} + V_{min.}}{V_{max.} - V_{min.}}$ (4) $m = \frac{V_{max.} - V_{min.}}{V_{max.} + V_{min.}}$

6. The AM wave contains three frequencies, viz :
(1) $\frac{f_c}{2}, \frac{f_c + f_s}{2}, \frac{f_c - f_s}{2}$ (2) $2f_c$, $2(f_c + f_s)$, $2(f_c - f_s)$
(3) f_c,$(f_c + f_s)$, $(f_c - f_s)$ (4) f_c , f_c, f_c

7. Which of the following is/are the limitations of amplitude modulation?
(1) Clear reception (2) High efficiency
(3) Small operating range (4) Good audio quality

8. In which of the region of earth's atmosphere temperature decreases with height?
(1) Ionosphere (2) Stratosphere (3) Troposphere (4) Mesosphere

9. In an amplitude modulated wave, for audio frequency of 500 cps, the appropriate carrier frequency will be :
(1) 50 c/s (2) 100 c/s (3) 500 c/s (4) 50000 c/s

10. In A.M., the total modulation index should not exceed one or else :
(1) the system will fail (2) distortion will result
(3) amplifier will be damaged (4) none of the above

11. An 'antenna' is :
(1) inductive (2) capacitive
(3) resistive above its resonance frequency (4) none of the above

12. The Q of a resonant transimission line is

(1) $Q = \frac{\omega}{LR}$ (2) $Q = \frac{\omega R}{L}$ (3) $Q = \frac{L}{R}$ (4) $Q = \frac{\omega L}{R}$

13. Range of frequencies allotted for commercial FM radio broadcast is -

(1) 88 to 108 MHz (2) 88 to 108 kHz (3) 8 to 88 MHz (4) 88 to 108 GHz

14. Which of the following is not transducer?

(1) Loudspeaker (2) Amplifier (3) Microphone (4) All of these

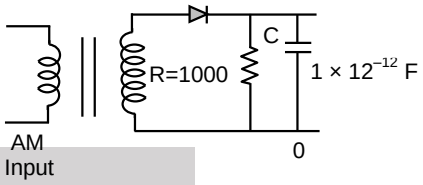
15. In the given detector circuit, the suitable value of carrier frequency is

(1) $<<10^9$ Hz

(2) $<<10^5$ Hz

(3) $>> 10^9$ Hz

(4) None of these



16. Amplitude modulation has

(1) One carrier (2) One carrier with high frequency

(3) One carrier with two side band frequencies (4) One carrier with infinite frequencies

17. Through which mode of propagation, the radio waves can be sent from one place to another

(1) Ground wave propagation (2) Sky wave propagation

(3) Space wave propagation (4) All of them

18. Sinusoidal carrier voltage of frequency 1.5 MHz and amplitude 50 V is amplitude modulated by sinusoidal voltage of frequency 10 kHz producing 50% modulation. The lower and upper side-band frequencies in kHz are

(1) 1490, 1510 (2) 1510, 1490 (3) $\frac{1}{1490}, \frac{1}{1510}$ (4) $\frac{1}{1510}, \frac{1}{1490}$

19. In an amplitude modulation with modulation index 0.5, the ratio of the amplitude of the carrier wave to that of the side band in the modulated wave is

(1) 4 : 1 (2) 1 : 1 (3) 1 : 2 (4) 2 : 1

20. A carrier frequency of 1MHz and peak value of 10V is amplitude modulated with a signal frequency of 10kHz with peak value of 0.5 V. Then the modulation index and the side band frequencies respectively are

(1) 0.05 and 1 ± 0.010 MHz (2) 0.5 and 1 ± 0.010 MHz

(3) 0.05 and 1 ± 0.005 MHz (4) 0.5 and 1 ± 0.005 MHz

21. In amplitude modulation, the bandwidth is

(1) Twice the audio signal frequency (2) Thrice the audio signal frequency

(3) Thrice the carrier wave frequency (4) Twice the carrier wave frequency

PART-II: NUMERICAL VALUE QUESTIONS

1. For a carrier frequency of 100 kHz and a modulating frequency of 5 kHz, the width of AM transmission in kHz is:

2. A transmitter supplies 9 kW to the aerial when unmodulated. The power radiated when modulated to 40% in kW is :

3. The antenna current of an AM transmitter is 8A when only carrier is sent but increases to 8.96 A when the carrier is sinusoidal modulated. The percentage modulations is -

4. The total power content of an AM wave is 900 W. For 100% modulation, the power transmitted by each side band in 10n W. Find the value of n :

5. 1000kHz carrier wave is amplitude modulated by the signal frequency 200 – 4000Hz. The channel width of this case in kHz is :

6. The area of the region covered by the TV broadcast by a TV tower of 100 m height is $n\pi \times 10^3$ km^2. Find n (radius of the earth = 6.4×10^6 m)

7. The distance of coverage of a transmitting antenna is 12.8 km. Then, the height of the antenna in m is (Given that radius of earth = 6400 km)

8. If the maximum amplitude of an amplitude modulated wave is 25 V and the minimum amplitude is 5 V, the modulation index is n/3. Find n -

Exercise-2

✍ Marked Questions can be used as Revision Questions.

*** Marked Questions may have more than one correct option.**

PART - I : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. This question has Statement –1 and Statement –2. Of the four choices given after the statements, choose the one that best describes the two statements. **[AIEEE - 2011, 4/120, –1]**

Statement –1

Sky wave signals are used for long distance radio communication. These signals are in general, less stable than ground wave signals.

Statement –2 :

The state of ionosphere varies from hour to hour, day to day and season to season.

(1) Statement –1 is true, statement –2 is false.

(2) Statement –1 is true, Statement –2 is true, Statement –2 is the correct explanation of Statement –1

(3) Statement –1 is true, Statement –2 is true, Statement–2 is not the correct explanation of Statement–1

(4) Statement–1 is false, Statement –2 is true

2. Which of the following four alternatives is not correct? **[AIEEE 2011, 11 May; 4/120, –1]**

We need modulation:

(1) to reduce the time lag between transmission and reception of the information signal

(2) to reduce the size of antenna

(3) to reduce the e fractional band width, that is the ratio of the signal band width to the centre frequency

(4) to increase the selectivity.

3.✍ A radar has a power of 1kW and is operating at a frequency of 10 GHz. It is located on a mountain top of height 500m. The maximum distance upto which it can detect object located on the surface of the earth (Radius of earth = 6.4×10^6 m) is : **[AIEEE - 2012, 4/120, –1]**

(1) 80 km (2) 16 km (3) 40 km (4) 64 km

4. A diode detector is used to detect an amplitude modulated wave of 60% modulation by using a condenser of capacity 250 pico farad in parallel with a load resistance 100 kilo ohm. Find the maximum modulated frequency which could be detected by it. **[JEE (Main) - 2013; 4/120, –1]**

(1) 10.62 MHz (2) 10.62 kHz (3) 5.31 MHz (4) 5.31 kHz

5. If a carrier wave c(t) = A sinω_ct, were to be amplitude modulated by a modulating signal m(t) = A sinω_mt, the equation representing the modulated signal [C_m(t)], and its modulatgion index, would be respectively :

(1) $C_m(t) = A(1 + \sin\omega_c t)\sin\omega_m t$ and 1 (2) $C_m(t) = A(1 + \sin\omega_c t)\sin\omega_m t$ and 2

(3) $C_m(t) = A(1 + \sin\omega_m t)\sin\omega_c t$ and 1 (4) $C_m(t) = A(1 + \sin\omega_m t)\sin\omega_c t$ and 2

6. For sky wave propagation, the radio waves must have a frequency range in between : **[JEE (Main) 2014_ONLINE TEST]**

(1) 1 MHz to 2 MHz (2) 45 MHz to 50 MHz

(3) 35 MHz to 40 MHz (4) 5 MHz to 25 MHz

7. Long range radio transmission is possible when the radiowaves are reflected from the ionosphere. For this to happen the frequency of the radiowaves must be in the range : **[JEE (Main) 2014_ Online Test]**

(1) 8–25 MHz (2) 80–150 MHz (3)150–500 kHz (4) 1–3 MHz

8. A signal of 5 kHZ frequency is amplitude modulated on a carrier wave of frequency 2 mHz. The frequencies of the resultant signal is/are : **[JEE(Main)-2015; 4/120, –1]**

(1) 2 MHz only (2) 2005 kHz, and 1995 kHz

(3) 2005 kHz, 2000 kHz and 1995 kHz (4) 2000 kHz and 1995 kHz

9. Choose the correct statement : **[JEE(Main)-2016; 4/120, –1]**

(1) In amplitude modulation the frequency of high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal

(2) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal.

(3) In frequency modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the frequency of the audio signal

(4) In amplitude modulation the amplitude of the high frequency carrier wave is made to vary in proportion to the amplitude of the audio signal

10. In amplitude modulation, sinusoidal carrier frequency used is denoted by ω_c and the signal frequency is denoted by ω_m. The bandwidth ($\Delta\omega_m$) of the signal is such that $\Delta\omega_m << \omega_c$. Which of the following frequency is not contained in the modulated wave ? **[JEE (Main) 2017; 4/120, –1]**

(1) $\omega_c - \omega_m$ (2) ω_m (3) ω_c (4) $\omega_m + \omega_c$

11. A telephonic communication service is working at carrier frequency of 10 GHz. Only 10% of it is utilized for transmission. How many telephonic channels can be transmitted simultaneously if each channel requires a bandwidth of 5 kHz? **[JEE(Main)-2018, 4/120, –1]**

(1) 2×10^5 (2) 2×10^6 (3) 2×10^3 (4) 2×10^4

12. In a communication system operating at wavelength 800 nm, only one percent of source frequency is available as signal bandwidth. The number of channels accommodated for transmitting TV signals of band width 6 MHz are (Take velocity of light $c = 3 \times 10^8$ m/s, $h = 6.6 \times 10^{-34}$ J-s] **[JEE(Main)-2019_09-01-2019_Shift-2]**

(1) 3.75×10^6 (2) 3.86×10^6 (3) 6.25×10^5 (4) 4.87×10^5

13. A TV transmission tower has a height of 140 m and the height of the receiving antenna is 40 m. What is the maximum distance upto which signals can be broadcasted from this tower in LOS (Line of sight) mode ? (Given : radius of earth = 6.4×10^6 m) **[JEE(Main)-2019_10-01-2019_Shift-1]**

(1) 80 km (2) 40 km (3) 48 km (4) 65 km

14. The modulation frequency of an AM radio station is 250 kHz, which is 10% of the carrier wave. If another AM station approaches you for licence what broadcast frequency will you allot ?

[JEE(Main)-2019_10-01-2019_Shift-2]

(1) 2750 kHz (2) 2250 kHz (3) 2900 kHz (4) 2000 kHz

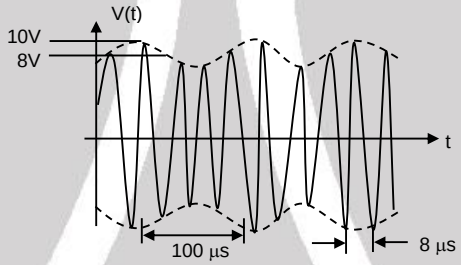
15. An amplitude modulated signal is given by $V(t) = 10\,[1+ 0.3\cos(2.2 \times 10^4 t)\sin(5.5 \times 10^5 t)]$. Here t is in seconds. The sideband frequencies (in kHz) are, [Given $\pi = 22/7$]

[JEE(Main)-2019_11-01-2019_Shift-1]

(1) 892.5 and 857.5 (2) 89.25 and 85.75

(3) 1785 and 1715 (4) 178.5 and 171.5

16. An amplitude modulated signal is plotted below : **[JEE(Main)-2019_11-01-2019_Shift-2]**



Which one of the following best describes the above signal ?

(1) $(9 + \sin(2.5\pi \times 10^5 t))\sin(2\pi \times 10^4 t)$ V

(2) $(1 + 9\sin(2\pi \times 10^4 t))\sin(2.5\pi \times 10^5 t)$V

(3) $(9 + \sin(2\pi \times 10^4 t))\sin(2.5\pi \times 10^5 t)$V

(4) $(9 + \sin(4\pi \times 10^4 t))\sin(5\pi \times 10^5 t)$V

17. A 100 V carrier wave is made to vary between 160 V and 40 V by a modulating signal. What is the modulation index ? **[JEE(Main)-2019_12-01-2019_Shift-1]**

(1) 0.4 (2) 0.6 (3) 0.5 (4) 0.3

18. The wavelength of the carrier waves in a modern optical fiber communication network is closed to :

[JEE(Main)-2019_08-04-2019_Shift-1]

(1) 1500 nm (2) 2400 nm (3) 600 nm (4) 900 nm

19. A signal $A\cos\omega t$ is transmitted using $\upsilon_0\sin\omega_0 t$ as carrier wave. The correct amplitude modulated (AM) signal is : **[JEE(Main)-2019_09-04-2019_Shift-1]**

(1) $\upsilon_0 \sin w_0 t + A\cos\omega t$

(2) $\upsilon_0 \sin\omega_0 t + \frac{A}{2}\sin(\omega_0 - \omega)t + \frac{A}{2}\sin(\omega_0 + \omega)t$

(3) $\upsilon_0\sin[\omega 0(1 + 0.01A\sin\omega t)t]$

(4) $(\upsilon_0 + A)\cos\omega t \sin\omega_0 t$

20. Given below in the left column are different modes of communication using the kinds of waves given in the right column. **[JEE(Main)-2019_10-04-2019_Shift-1]**

A.	Optical Fibre Communication	P.	Ultrasound
B.	Radar	Q.	Infrared Light
C.	Sonar	R.	Microwaves
D.	Mobile Phones	S.	Radio Waves

From the options given below, find the most appropriate match between entries in the left and the right column.

(1) A - Q, B-S, C-P, D-R (2) A-S, B-Q,C-R, D-P

(3) A-Q, B-S, C-R, D-P (4) A-R, B-P, C-S, D-Q

21. A message signal of frequency 100 MHz and peak voltage 100 V is used to execute amplitude modulation on a carrier wave of frequency 300 GHz and peak voltage 400 V. The modulation index and difference between the two side band frequencies are : **[JEE(Main)-2019_10-04-2019_Shift-1]**

(1) 4 ; 1×10^8 Hz (2) 0.25 ; 1×10^8 Hz

(3) 0.25 ; 2×10^8 Hz (4) 4 ; 2×10^8 Hz

Answers

EXERCISE # 1

PART-I

1.	(3)	**2.**	(1)	**3.**	(3)
4.	(2)	**5.**	(4)	**6.**	(3)
7.	(3)	**8.**	(3)	**9.**	(4)
10.	(2)	**11.**	(1)	**12.**	(3)
13.	(1)	**14.**	(2)	**15.**	(1)
16.	(3)	**17.**	(4)	**18.**	(1)
19.	(1)	**20.**	(1)	**21.**	(1)

PART-II

1.	10.00	**2.**	09.72	**3.**	71.00
4.	15.00	**5.**	08.00	**6.**	01.28
7.	12.80	**8.**	02.00		

EXERCISE # 2

PART - I

1.	(4)	**2.**	(1)	**3.**	(1)
4.	(2)	**5.**	(3)	**6.**	(3)
7.	(1)	**8.**	(3)	**9.**	(4)
10.	(2)	**11.**	(1)	**12.**	(3)
13.	(4)	**14.**	(4)	**15.**	(2)
16.	(3)	**17.**	(2)	**18.**	(1)
19.	(2)	**20.**	(2)	**21.**	(3)